Odin OS: Cryptographically Authorized Memory Mutation in a RISC-V PMP-Enforced Agent Execution Environment

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Abstract

Current paradigms in autonomous artificial intelligence rely predominantly on probabilistic software guardrails—natural language prompts, output schemas, and user-space virtual sandboxes—to constrain agent behavior. These mechanisms are vulnerable to direct prompt injection, semantic jailbreaks, and memory-state corruption. In this paper, we present **Odin OS**, a bare-metal, zero-allocation (`#![no_std]`) 64-bit RISC-V microkernel that anchors autonomous agent trust boundaries in processor Physical Memory Protection (PMP) comparators. Operating on the central axiom that **state which cannot prove its right to exist cannot be committed to memory or disk**, Odin OS enforces capability isolation via cache-line aligned (256-byte) binary tokens termed **TransitionDeeds**. We establish the primary falsifiable claim: **an untrusted U-Mode agent executing arbitrary instructions cannot commit a store to physical memory outside its explicitly authorized TransitionDeed window or after lease revocation without triggering an architectural CPU access fault (`mcause = 7`) that aborts the write and leaves target memory unmodified.**

We present seven core contributions: (1) **Cryptographic Authorization**: Integration of RFC 8032 Ed25519 digital signature verification in pure `#![no_std]` without heap allocation (`no_alloc`), bound to a 32-byte pinned root public key, with decoupled Policy Decision Point (PDP) authority isolation; (2) **Bi-Directional $W \oplus X$ Privilege Configuration**: Domain partitioning configuring user code as read-execute (`R-X`, store attempts fault with `mcause = 7`), while user stack memory is read-write non-executable (`RW-`, instruction fetch attempts fault with `mcause = 1` and are safely redirected to caller return addresses); (3) **Dynamic Capability Revocation**: Privileged syscall interface (`SYS_REVOKE_LEASE`, `ecall 13`) that dynamically collapses PMP comparator windows to `OFF`; (4) **17-Case Adversarial Matrix Evaluation**: Bare-metal QEMU verification demonstrating that all 17 adversarial test cases passed under emulation, comprising processor-enforced access faults and software-enforced authorization checks; (5) **Machine-Checked Formal Properties**: Z3 SMT models (`QF_BV`) verifying bounded integer overflow immunity, memory disjointness, and anti-replay sequence monotonicity; (6) **Mutation Testing Sensitivity**: Controlled injection of 15 architectural and cryptographic mutations, with 15 out of 15 detected and killed by the test suite; and (7) **End-to-End LLM Bridge**: Live coupling to local `google/gemma-4-e4b` inference, demonstrating that authorized memory stores execute under configured PMP permissions without per-store kernel mediation, while out-of-bounds mutation attempts are rejected at the PDP or trapped by processor PMP checks with no modification to inspected memory locations.

1 Introduction & Problem Statement

The rapid deployment of autonomous AI agents tasked with executing arbitrary code, modifying datastores, and manipulating abstract syntax trees (ASTs) has exposed an architectural vulnerability in modern computing. Contemporary agent runtimes treat safety as an inference-time alignment problem, attempting to prevent unauthorized state transitions through prompt engineering, output schemas, and container sandboxes.

This premise fails against adversarial attacks for three fundamental reasons:

1. **Prompt Invariance Failure**: No natural language instruction or heuristic system prompt is invariant against adversarial prompt injection. An adversarial token sequence can reliably cause a neural model to ignore its operational parameters.
2. **Semantic Escape**: In traditional operating systems, a compromised or hallucinating agent process running in user space possesses write access to its entire virtual address space. Once a store instruction is issued to the CPU, the underlying memory controller commits the mutated bytes regardless of agent intent.
3. **Post-Hoc Remediation**: Existing detection systems operate **after** memory has been mutated, relying on logging, rollbacks, or audit trails that incur latency and fail to prevent state corruption.

To resolve this vulnerability, we invert the trust model. Rather than attempting to guarantee that neural models will never produce malicious or errant store instructions, we construct an execution substrate where processor Physical Memory Protection restricts U-Mode accesses and raises architectural access faults when configured permissions are violated.

2 Threat Model & Trust Assumptions

2.1 The Central Axiom & Security Invariant

Odin OS is constructed around a single deterministic axiom:

> ***"State that cannot prove its right to exist cannot be committed to memory or disk."***

This enforces the fundamental security invariant:

$$\text{Agent Request} \neq \text{Authority Grant}$$

2.2 Trust Boundary Classification

Components are strictly categorized across five trust boundaries:

- **Policy Decision Point (PDP)**: **Trusted Authority**. Evaluates agent mutation requests against declarative security policies, manages monotonic sequence counters, and holds the root Ed25519 signing key. Operates in two modes: Mode A (Deterministic Research Fixture) using a documented test key for reproducible evaluation, and Mode B (Independently Provisioned Authority) with an external CSPRNG seed.
- **CPU Hardware (PMP Comparators / CSRs)**: **Trusted Enforcer**. Enforces address comparator matches, raises access faults, and enforces execution privilege modes.
- **M-Mode Microkernel (TCB)**: **Trusted Verifier & Enforcer**. Verifies Ed25519 signatures against the pinned root public key, programs PMP registers, and controls monotonic replay sequences.
- **Autonomous Agent (U-Mode Task)**: **Untrusted (Adversary)**. Executes untrusted model instructions and arbitrary arithmetic. Possesses zero signing keys and zero capability to modify PMP registers.
- **Host / Python Bridge**: **Untrusted Transport Channel**. Transports model tokens and capability requests; possesses zero private keys and is assumed to be completely compromiseable.

2.3 Enforcement Boundary: Spatial PEP vs. Semantic PDP

A critical architectural boundary governs Odin OS: the formal distinction between Policy Enforcement (PEP) and Policy Decision (PDP):

- **Spatial & Temporal PEP (Microkernel & PMP)**: Odin OS operates strictly as a hardware-enforced Policy Enforcement Point. It guarantees spatial memory confinement (stores cannot retire outside authorized byte spans) and temporal containment (stores cannot occur before lease entry or after revocation). Crucially, the PEP is agnostic to higher-level business or application semantics: it enforces physical capability constraints, not whether a proposed mutation makes contextual sense.
- **Semantic PDP (Host Authority)**: The Policy Decision Point evaluates declarative domain policies (e.g., agent identity, AST anchor legitimacy, allowed opcodes, lease duration caps) prior to minting and signing a TransitionDeed.
- **Application Semantic Correctness**: Establishing whether an authorized business action is contextually “appropriate” (e.g., high-level semantic truth, application invariants) is explicitly decoupled from the microkernel’s TCB. If an agent is tricked into requesting an unauthorized offset, the PDP denies the deed; if an agent attempts a store without or beyond an active deed, the CPU PMP traps the access.

3 TransitionDeed Protocol

State transitions in Odin OS are mediated through **TransitionDeeds**. A TransitionDeed is a standardized, 256-byte physical binary packet designed for cache-line alignment with zero heap allocations.

Offset	Field	Type	Description
0x00..0x02	magic	[u8; 2]	Magic identifier 0x59 0x47 (“YG”)
0x02..0x03	version	u8	Protocol specification version (0x02)
0x03..0x04	flags	u8	Capability flags (MUTATION, AST_STRICT)
0x04..0x08	sequence	u32	Monotonically increasing deed nonce
0x08..0x18	issuer_did	[u8; 16]	Cryptographic identity of the issuer
0x18..0x28	target_urn	[u8; 16]	Target entity uniform resource name
0x28..0x48	prev_deed_hash	[u8; 32]	Cryptographic hash of prior deed
0x48..0x68	base_digest	[u8; 32]	SHA-256 pre-mutation state digest
0x68..0x6C	lease_cycles	u32	Execution cycle lifetime bound
0x6C..0x70	operation	u8 + u8 + u16	Opcode, escalation level, reserved
0x70..0x74	anchor_id	u32	Target AST Anchor identifier
0x74..0x78	target_offset	u32	Authorized buffer start offset
0x78..0x7C	target_length	u32	Authorized buffer span length
0x7C..0x9C	new_digest	[u8; 32]	Expected post-mutation state digest
0x9C..0xDC	signature	[u8; 64]	RFC 8032 Ed25519 digital signature
0xDC..0x100	padding	[u8; 36]	Zero padding to align to 256 bytes

Table 1: Physical layout of the 256-byte cache-line aligned TransitionDeed structure.

4 Kernel Architecture & PMP Enforcement

4.1 Privilege Domain Layout

Odin OS partitions physical memory into five isolated privilege domains:

Domain	Physical Span	Mode	PMP Mode	Privilege & Architectural Purpose
Domain 0	0x10000000 .. 0x10001000	M-Mode	Default / ---	UART0 MMIO (Revoked from U-Mode)
Domain 1	0x80000000 .. 0x80010000	M-Mode	Default / ---	Privileged Kernel: Code, Stack, Keys, Traps
Domain 2	0x80010000 .. 0x80018000	U-Mode	TOR / R-X	Untrusted Agent Task Code ($W \oplus X$)
Domain 3	0x80018000 .. 0x80020000	U-Mode	TOR / RW-	Untrusted Agent Stack (Non-Executable)
Domain 4	0x80020000 .. lease_start	None	TOR / ---	Gated Pre-Lease AST Boundary
Domain 5	lease_start .. lease_end	U-Mode	TOR / RW-	Explicitly Authorized Transition-Deed Slice
Perimeter	All other physical addresses	None	OFF / ---	Processor Default Deny (No Access)

Table 2: Hardened 7-Region PMP Layout and Privilege Partitioning.

4.2 Hardware W^X Stack Execution Prevention & Trap Handling

When user mode attempts to execute code from the user stack (`0x80018100`), the CPU instruction fetch unit triggers an Instruction Access Fault (`mcause = 1`). The M-Mode trap handler intercepts the fault, records the invariant violation, and safely redirects `mepc` back to the return address `frame.ra` in `.user_text`:

```
if mcause == 1 && mepc >= USER_STACK_START as u64 && mepc < USER_STACK_END as u64 {
    unsafe { core::arch::asm!("csrw mepc, {0}", in(reg) frame.ra); }
    return;
}
```

This guarantees that stack execution attempts cannot escalate privileges or enter infinite fault loops.

4.3 Temporal Lease Revocation (`SYS_REVOKE_LEASE`)

To prevent temporal capability leakage, Odin OS introduces microkernel syscall 13 (`sys_revoke_lease`). When invoked, M-Mode clears active lease bounds and sets PMP Entry 4 to `OFF`. Subsequent store attempts to the previously valid lease slice immediately trigger an architectural Store Access Fault (`mcause = 7`).

5 Cryptographic Authorization & Authority Isolation

5.1 Independent Policy Decision Point (PDP) Architecture

In compliance with the invariant $\text{Agent Request} \neq \text{Authority Grant}$, Odin OS decouples agent token generation from cryptographic signing. The private signing key is isolated within an independent Policy Decision Point service (`pdp_authority_service.py`). Neither the untrusted agent nor the communication bridge possesses access to private key material.

The PDP supports two operational modes:

1. **Mode A (Deterministic Research Fixture)**: Employs a public, documented test key (`b"odin_root_authority_seed_v2_key!"`) ensuring offline, deterministic test reproducibility.
2. **Mode B (Independently Provisioned Authority)**: Loads an external 32-byte CSPRNG seed stored outside the repository and injected into the microkernel at compile time via `build.rs` (`ODIN_ROOT_PUBKEY_HEX`). Mode B fails closed if key material is missing or invalid.

When an agent proposes a state mutation, the request is evaluated against a default-deny policy registry enforcing: identity verification, AST anchor whitelisting, spatial lease clamping, temporal caps, and monotonic sequence allocation. Only approved requests receive signatures. In empirical testing across 12 attack vectors (unauthorized anchors, invalid opcodes, oversized mutations, excessive lease durations, rogue agent identities, out-of-bounds offsets, zero-length mutations, post-sign tampering, sequence restarts, and Mode B key separation), the PDP rejected all 12 evaluated adversarial cases.

5.2 Canonical Domain Separation & Zero-Allocation Verifier

The signed message encompasses 172 contiguous bytes:

$$\text{Message} = \text{DOMAIN_SEPARATOR}_{16\text{B}} \,\|\, \text{Metadata}_{156\text{B}}$$

where `DOMAIN_SEPARATOR` is `"ODIN_DEED_V2\0\0\0\0"`. Signatures are verified against a 32-byte public key pinned in kernel read-only memory (`TRUSTED_ROOT_PUBLIC_KEY`). The verifier executes via `ed25519-compact` in pure `#![no_std]` with zero heap allocations.

6 Experimental Methodology

Our experimental evaluation executes on a 64-bit RISC-V platform (`riscv64gc-unknown-none-elf`) under QEMU 11.1.3 `virt`. Evidence is audited via a two-layer verification pipeline:

- **Layer A (In-Kernel)**: Direct M-Mode observation of trap causes, probed addresses, readback values, and memory preservation.
- **Layer B (External Host Verifier)**: An independent Python verifier (`tools/verify_evidence.py`) that strictly validates receipt schemas, enforces monotonic test sequences (1..17), and re-evaluates all invariant assertions independently of the kernel's claimed status.

6.1 Claim-to-Evidence Mapping

To provide peer reviewers with an explicit, bounded starting point for falsification, Table 3 maps each core architectural property to its trusted components, empirical test artifacts, and explicit non-scope boundaries:

ID	Claim / Property	TCB & Assumptions	Supporting Artifact / Test	Explicitly Outside Scope
C1	Spatial Confinement	RISC-V PMP comparators, M-Mode trap handler	Adversarial Tests 1–4, 9, 10; Z3 Proof 1	Timing side-channels; non-atomic split stores on unverified silicon
C2	Temporal Revocation	M-Mode syscall 13 (SYS_REVOKE_LEASE)	Adversarial Test 16; Mutation M09	Autonomous timer-interrupt cycle preemption (mcycle)
C3	Cryptographic Authority	ed25519-compact (no_std), pinned root key	Adversarial Tests 11, 13, 14; Mutations M01, M02, M11–M13	Host signing key compromise outside the microkernel
C4	Anti-Replay Nonce	In-kernel monotonic sequence counter last_seq	Adversarial Test 12; Mutation M03; Z3 Proof 2	NVRAM counter rollback across reset without re-keying
C5	Bi-directional $W \oplus X$	PMP Entry 1 (R-X), Entry 2 (RW-)	Adversarial Tests 7 & 8; Mutations M06, M15	Return-Oriented Programming (ROP) in valid user text
C6	Semantic Appropriateness	External Policy Decision Point (PDP)	PDP Test Suite (12 attack vectors rejected)	**Outside Odin OS PEP scope.** Kernel enforces capability bounds, not business logic.

Table 3: Claim-to-evidence mapping across TCB assumptions, empirical artifacts, and non-goals.

7 Adversarial Results

#	Test Invariant Case	Target Addr	Width	Exp Fault	Obs Fault	mcause	Status
1	Exact Lower Bound	0x80020040	1B	No	No	0x00	**PASS**
2	Exact Upper Bound - 1	0x8002005F	1B	No	No	0x00	**PASS**
3	One Byte Below Lease	0x8002003F	1B	Fault	Fault	0x07	**PASS**
4	One Byte Above Lease	0x80020060	1B	Fault	Fault	0x07	**PASS**
5	Kernel Code Write	0x80000000	1B	Fault	Fault	0x07	**PASS**
6	Kernel Stack Write	0x8000FFF8	1B	Fault	Fault	0x07	**PASS**
7	User Code Write ($W \oplus X$)	0x80010000	1B	Fault	Fault	0x07	**PASS**
8	User Stack Execute ($W \oplus X$)	0x80018100	0B	Fault	Fault	0x01	**PASS**
9	Unmapped Perimeter Write	0x80030000	1B	Fault	Fault	0x07	**PASS**
10	Multi-byte Overlap Crossing	0x8002005E	4B	Fault	Fault	0x07	**PASS**
11	Forged Deed Issuer DID	0x00000000	0B	Reject	Reject	Syscall / 1	**PASS**
12	Replayed Deed Sequence	0x00000000	0B	Reject	Reject	Syscall / 2	**PASS**
13	Forged Deed Invalid Ed25519	0x00000000	0B	Reject	Reject	Syscall / 4	**PASS**
14	Deed Tampered Metadata Ed25519	0x00000000	0B	Reject	Reject	Syscall / 4	**PASS**
15	U-Mode CSR Access pmpcfg0	0x00000000	0B	Fault	Fault	0x02	**PASS**
16	Expired Lease Revocation	0x80020040	1B	Fault	Fault	0x07	**PASS**
17	U-Mode UART0 MMIO Write	0x10000000	1B	Fault	Fault	0x07	**PASS**

Table 4: Empirical receipts from the 17-case adversarial verification matrix evaluated under QEMU (17/17 PASS).

7.1 Architectural Analysis of Multi-Byte Overlap Crossing (Test 10)

A 4-byte store at 0x8002005E straddles the lease boundary (0x80020060), with bytes [0x5E..0x5F] inside the lease and [0x60..0x61] outside. Because 0x8002005E is 2-byte aligned but not 4-byte aligned ($0x5E \neq 0 \bmod 4$), this instruction tests hardware behavior across an asymmetric protection boundary.

Under the **RISC-V Unprivileged Architecture Specification (Section 2.6)**:

"Misaligned memory accesses are not guaranteed to be atomic."

In our QEMU 11.1.3 virt evaluation, the processor performs address checking across split memory accesses prior to retiring the store, raising a Store Access Fault (mcause = 7) and leaving target memory unmodified (0x00). We verified that an unaligned 4-byte store entirely within the lease (0x80020052) succeeds without fault, demonstrating that QEMU supports unaligned accesses and that the fault at 0x8002005E is triggered solely by the PMP boundary violation.

Physical Silicon Implication: While software simulation in QEMU enforces all-or-nothing rollback on this split access, physical RISC-V microarchitectures are not architecturally bound by the base ISA to guarantee atomicity for misaligned stores. A physical core may either trap with `mcause = 6` (Store Address Misaligned) or commit the lower bytes before triggering a PMP fault on the upper half. Consequently, Odin OS establishes the architectural requirement that robust production deployments must couple PMP hardware fences with software-level staging buffers (shadow ASTs) to ensure strict transaction atomicity across arbitrary silicon implementations.

8 Mutation Testing

To evaluate test suite sensitivity, we injected 15 deliberate architectural and cryptographic faults:

ID	Injected Architectural Mutation	Detecting Invariant Test	Result
M01	Bypass issuer DID check	Test 11 (FORGED_DEED_UNKNOWN_ISSUER)	KILLED
M02	Disable Ed25519 signature verification	Tests 13 & 14 (FORGED_DEED_ED25519)	KILLED
M03	Permit sequence reuse (< instead of <=)	Test 12 (REPLAYED_DEED_SEQUENCE)	KILLED
M04	Wrapping addition on bounds (disable checked)	prop_deed_overflow_prevention	KILLED
M05	Expand PMP lease by 1 byte in setup_pmp	Test 4 (ONE_BYTE_ABOVE_HALT)	KILLED
M06	Configure user code as RWX (0x0F)	Test 7 (USER_CODE_WRITE_HALT)	KILLED
M07	Map UART MMIO in U-Mode	Test 17 (U_MODE_UART_MMIO_WRITE)	KILLED
M08	Corrupt register restore in trap_entry	Architectural Register Invariants	KILLED
M09	Skip PMP reprogramming on lease revocation	Test 16 (EXPIRED_LEASE_REVOCATION)	KILLED
M10	Tamper with receipt status emission	Dynamic Invariant Assertions	KILLED
M11	Mutate domain separator string	Host Test 8 (Cross-protocol)	KILLED
M12	Truncate canonical payload slice	Host Test 20 (Valid signature)	KILLED
M13	Corrupt pinned root public key	Matrix Test 1 & Host Test 20	KILLED
M14	Inject forged PASS into external Layer B trace	External Verifier (verify_evidence.py)	KILLED
M15	Omit stack execution trap redirection	Matrix Test 8 (Infinite loop)	KILLED

Table 5: Controlled mutation testing results demonstrating detection sensitivity across 15 injected faults (15/15 killed).

9 Formal Assurance

Using the Z3 Theorem Prover version 5.3.1 (`QF_BV`), we verified five machine-checked properties of the formal SMT model:

1. **Integer Overflow Bound**: Verified that under the model's assumptions, checked addition guarantees $\text{end_offset} > \text{target_offset}$ (`unsat`).
2. **Modeled Memory Disjointness**: Verified that for modeled address ranges, an accepted capability lease is disjoint from modeled privileged addresses (`unsat`).
3. **Modeled Buffer Containment**: Verified that modeled accepted leases remain within modeled buffer bounds (`unsat`).
4. **Identical Sequence Non-Replay**: Verified that resubmission of an identical sequence value is unsatisfiable under strict monotonicity (`unsat`).
5. **Stale Sequence Non-Replay**: Verified that any sequence number $s \leq \text{last_sequence}$ is unsatisfiable under strict monotonicity (`unsat`).

10 Limitations

1. **Simulation vs. Silicon**: All empirical measurements were conducted under QEMU 11.1.3 `virt`. Hardware characteristics such as non-zero grain size ($G \geq 4$), store pipeline write-buffer draining, and physical bus split-access behavior require physical silicon validation.
2. **Autonomous Timer Expiration**: Capability leases are revoked explicitly via microkernel syscalls (`sys_revoke_lease`); autonomous timer-interrupt cycle counting (`mcycle`) is designed for future work.
3. **Spatial vs. Semantic Integrity**: PMP enforces spatial write boundaries (`where` writes occur); verifying that RAM contents bitwise match cryptographic digests (`base_digest -> new_digest`) requires shadow AST staging buffers and commit barriers, specified as proposed architecture.

11 Related Work

- **seL4 Microkernel**: Formal verification of capability-based microkernel isolation. Odin OS targets fine-grained tokenized AST leases for autonomous AI agents directly over bare-metal RISC-V PMP.
- **CHERI**: Fat-pointer capability hardware extensions. Odin OS enforces capabilities on standard, unmodified commodity RISC-V silicon without requiring custom ISA extensions.
- **Memory Protection Keys (Intel MPK)**: Hardware memory domain toggling. Odin OS isolates capability registers in privileged Machine Mode (M-Mode), preventing unprivileged User Mode tampering.

12 Future Hardware Directions

Phase C of the Odin OS research roadmap targets physical deployment on physical silicon and reconfigurable logic platforms:

- **AMD Kria KV260 FPGA/MPSoC**: Dual-rail architecture integrating ARM Cortex-A53 application cores with programmable logic (PL) for hardware-accelerated AST graph traversal, direct AXI memory interlocks, and cryptographic digest hashing engines.
- **Physical RISC-V Silicon**: Deployment on 64-bit RISC-V application platforms (e.g., Microchip PolarFire SoC, StarFive JH7110), addressing store-buffer pipeline fence draining (`fence rw, rw`), non-volatile sequence persistence in secure EEPROM/NVRAM, and hardware timer interrupt cycle bounds (`mcycle`).

13 Conclusion

By combining zero-allocation Ed25519 signature verification, bi-directional $W \oplus X$ privilege configuration, temporal capability lease revocation, and evaluating the prototype against a 17-case QEMU adversarial matrix, 15 mutation tests, and machine-checked formal SMT models, Odin OS demonstrates an empirical architecture for constraining untrusted autonomous agent execution within hardware-enforced spatial boundaries.

14 Reproducibility & Artifact Availability

The complete Odin OS codebase, linker scripts, formal SMT specifications, and Layer B external verifiers are open-source and permanently archived on Zenodo.

- **Permanent Archive DOI**: `https://doi.org/10.5281/zenodo.23928302`
- **Zenodo Deposit**: `https://zenodo.org/records/23928302`
- **Source Repository**: `https://github.com/nwalker85/odinos`
- **Target Triple**: `riscv64gc-unknown-none-elf`
- **Licensing**: Apache-2.0 (Microkernel & Verification Tools); CC BY 4.0 (Manuscript)
- **External Verifier**: `python3 tools/verify_evidence.py`
- **Formal Proof Command**: `z3 formal/deed_bounds_verification.smt2 && z3 formal/sequence_anti_replay.smt2`